

K. Sai Charitha

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EDUCATION

VIT-AP University

B.Tech in Computer Science and Engineering

September 2023 – June 2027

Amaravati, Andhra Pradesh

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL

Web Technologies: HTML, CSS, JavaScript, RESTful APIs

Backend & AI: FastAPI, RAG, LLMs, Vector Embeddings, Prompt Engineering

Cloud & DevOps: AWS, Docker, Docker Compose, Kubernetes

Tools & APIs: Git, GitHub, Alembic, Razorpay API, VS Code

PROJECTS

Full-Stack Todo Application

[github](#)

FastAPI, React.js, PostgreSQL, Docker, Kubernetes, JWT, AES-256

- Built a secure full-stack to-do web app with enterprise-grade features including JWT authentication and Multi-Factor Authentication (MFA) via Google Authenticator.
- Implemented AES-256 encryption on all todo data to ensure end-to-end data security.
- Integrated Razorpay payment gateway for a premium subscription plan and added PDF export and statistics dashboard using Recharts and jsPDF.
- Containerised the application using Docker & Docker Compose and deployed on Kubernetes for scalable orchestration.

Campus RAG Assistant

[github](#)

Python ChromaDB, Groq (LLaMA 3.3 70B), Sentence-Transformers, Streamlit

- Built and deployed a Retrieval-Augmented Generation (RAG) system from scratch (without LangChain) to answer questions over college academic and placement documents with source citations.
- Designed a custom PDF ingestion pipeline with page-aware chunking and sentence-transformer embeddings, stored in ChromaDB for semantic retrieval.
- Implemented a relevance-threshold guardrail to prevent hallucinated answers on out-of-scope queries, and grounded generation using Groq-hosted LLaMA 3.3 70B.
- Built a custom 19-question evaluation suite achieving 89.5% retrieval accuracy, with documented failure analysis; deployed a live chat interface using Streamlit Community Cloud.

CERTIFICATIONS

IBM Generative AI Certificate

Covered prompt engineering, LLM fundamentals, and applied GenAI use cases

AWS Academy Cloud Architecting

Training in AWS cloud architecture, infrastructure design, and cloud best practices.